

Anita Petrovic

[LinkedIn](#) | Geneva, Switzerland | anita.petrovic@cern.ch | [Portfolio link](#)

Work Experience

CERN, Mechanical Engineer

September 2024 – Present

Cryogenics feasibility and conceptual design for the [Future Circular Collider \(FCC-ee\)](#)

Geneva, Switzerland

- Develop Python models of heat loads and helium flow in cryogenic distribution systems: conduction/radiation heat inleak, pressure drop, temperature/enthalpy evolution, mass-flow profiles, JT/flash behaviour, and node mixing to evaluate design options.
- Sized two-phase helium and rupture-disk line concepts for superconducting RF cryomodules through studies of vapour velocity, helium inventory and, level stability, flow velocity, and pressure differentials.
- Contributed to the FCC Feasibility Study Report (cryogenics).
- Define cryogenic requirements and integration constraints for utilities, electrical racks, building interconnections, and installation strategy across surface and underground infrastructure.
- Coordinate cryogenic design and integration activities with multidisciplinary engineering and infrastructure teams; serve as the CAD/PLM point of contact for cryogenics.

Lucid Motors, Chassis Mechanical Design Engineering Intern

May 2022 – August 2022

Luxury Electric Vehicle Company

Newark, CA

- Owned the design of a bushing fixture to test components of the Lucid Air suspension system.
- Used CATIA FEA to limit fixture deflection and verified that the design could withstand a maximum load of 18.2 kN.
- Created and executed a static and dynamic testing plan with the CTW LA-48 actuator.
- Wrote a MATLAB program to analyze test data from suspension bushings to verify static and dynamic stiffness performance.
- Designed, prototyped, and tested string potentiometer fixtures to collect and analyze rear steering data.
- Set up sensors and DAQ for these tests, using VBOX and CANalyzer for data logging.
- Developed a concept jig model for future rear steering testing of the Lucid Gravity SUV.

Tesla, Structural Design Engineering Intern

September 2021 – December 2021

Electric Vehicle and Clean Energy Company

Fremont, CA

- Owned the redesign of SEMI structural roof components, achieving a 5.4% mass reduction while addressing stamping feasibility, mass manufacturing, durability, and component integration.
- Independently designed and implemented new toe features for Model S/X, reducing cost and weight by over 7% while retaining structural integrity for frontal and corner crashes.
- Iterated designs for FDM-printed 3D fixtures with feedback from factory technicians that addressed NVH issues and reducing application time of adhesives by 10.5%.
- Updated GD&T for new product designs for Model S/X.

Publications and Talks

- Contributor, [FCC Feasibility Study Report, Vol. 2](#), Cryogenic Systems chapter (Eur. Phys. J. Spec. Top., 2025)
- Speaker, FCC Week 2025 and 2026: FCC-ee cryogenics status and baseline updates

Projects

EcoCar, Mechanical Team Lead

September 2018 – May 2023

Shell Eco-marathon, Fuel Cell Vehicle Team - University of Alberta

Edmonton, AB

- Led mechanical work for a fuel-cell vehicle team, managing 12 members, sponsorships, and a \$25,000 annual budget.
- Modeled the fuel cell, supercapacitor, DC/DC converter, and motor system in Simulink; designed a 2020 Urban-Concept wheel hub that achieved a 48% weight reduction.
- Managed a four-month fiberglass and carbon composite body layup process.
- Developed compact suspension/steering concepts for a confined monocoque.

Education

University of Alberta

June 2023

BSc Mechanical Engineering with Distinction

Edmonton, AB

- Scovan Engineering Prize in Design Project Management
- Final two academic years GPA: 3.9/4.0

Skills

CAD: CATIA V5/3DX, SOLIDWORKS (CSWP certification)

Analysis/Programming: Python, MATLAB, ANSYS, CATIA FEA, Excel

Engineering: Cryogenic systems, heat-transfer problems, thermal-fluid modelling, design for manufacturing, mechanical design, composite manufacturing, integration, experimental testing and validation, technical reporting, documentation